

DSC 140B

Representation Learning

Lecture 02 | Part 1

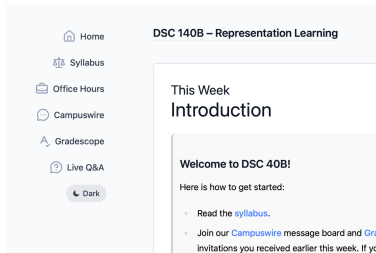
News

News

- ▶ **Homework 01 released**, due next Wednesday.
 - ▶ Optional!
 - ▶ Remember: must be handwritten.
- ▶ **Quiz 01 tonight**, 8pm (in this room).
 - ▶ Optional!

Attendance

- ▶ To earn credit for lecture attendance, you'll need to respond to (most of) the in-class polls.
- ▶ We'll use the “Live Q&A” on the course page:

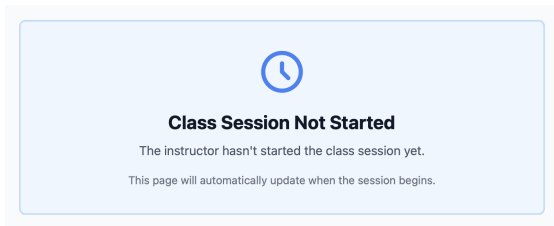


Important!

- ▶ To use the Live Q&A, you need to be on UCSD wifi.
 - ▶ If you're not, the page won't load.

Setup

- ▶ Go ahead and open the Live Q&A page now.
 - ▶ Remember, it is linked at `dsc140b.com`.
- ▶ You'll be asked for your PID. Enter it.
- ▶ You should then see:¹



¹If you couldn't get it to work, let me know after class.

Exercise

Do you plan on taking the quiz tonight?

- ▶ True = Yes
- ▶ False = No

By the way...

Ask

Ask a question

0 / 1000

Type your question here...

Submit Question

Tip: Questions are visible to your instructor in real-time.

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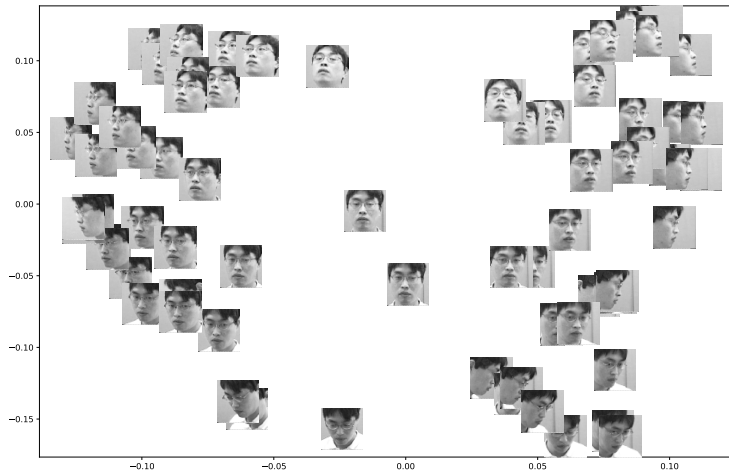
Lecture 02 | Part 2

Why Linear Algebra?

Last Time



Last Time



Dimensionality Reduction

- ▶ This is an example of **dimensionality reduction**:
 - ▶ Input: vectors in $\mathbb{R}^{10,000}$.
 - ▶ Output: vectors in $\mathbb{R}^2$.
- ▶ The method which produced this result is called **Laplacian Eigenmaps**.
- ▶ How does it work?

A Preview of Laplacian Eigenmaps

To reduce dimensionality from d to d' :

1. Create an undirected **similarity graph** G
 - ▶ Each vector in $\mathbb{R}^d$ becomes a node in the graph.
 - ▶ Make edge (u, v) if u and v are “close”
2. Form the **graph Laplacian matrix**, L :
 - ▶ Let A be the adjacency matrix, D be the degree matrix.
 - ▶ Define the graph Laplacian matrix, $L = D - A$.
3. Compute d' **eigenvectors** of L .
 - ▶ Each eigenvector gives one new feature.

Why eigenvectors?

- ▶ We will cover Laplacian Eigenmaps in much greater detail.
- ▶ For now: why do eigenvectors appear here?
 - ▶ What are eigenvectors?
 - ▶ How are they useful?
 - ▶ Why is linear algebra important in ML?

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Lecture 02 | Part 3

Coordinate Vectors

Coordinate Vectors

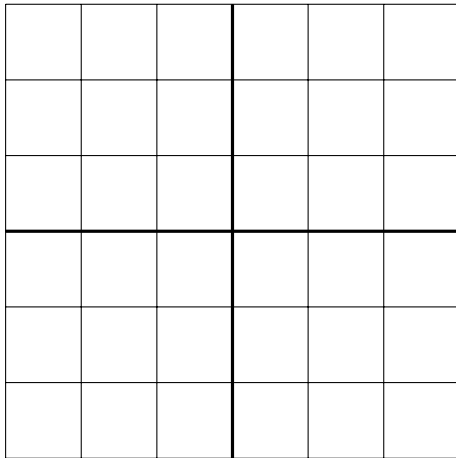
- We can write a vector $\vec{x} \in \mathbb{R}^d$ as a **coordinate vector**:

$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{pmatrix}$$

Example

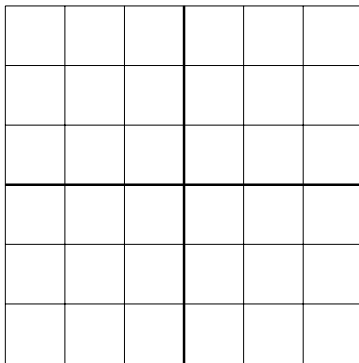
$$\vec{x} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

$$\vec{y} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$



Standard Basis

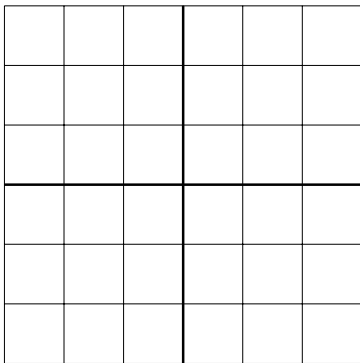
- ▶ Writing a vector in coordinate form requires choosing a **basis**.
- ▶ The “default” is the **standard basis**: $\hat{e}^{(1)}, \dots, \hat{e}^{(d)}$.



Standard Basis

- When we write $\vec{x} = (x_1, \dots, x_d)^T$, we mean that $\vec{x} = x_1 \hat{e}^{(1)} + x_2 \hat{e}^{(2)} + \dots x_d \hat{e}^{(d)}$.

Example: $\vec{x} = (3, -2)^T$



Standard Basis Coordinates

- In coordinate form:

$$\hat{e}^{(i)} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \\ 0 \end{pmatrix}$$

where the 1 appears in the i th place.

Exercise

Let $\vec{x} = (3, 7, 2, -5)^T$. What is $\vec{x} \cdot \hat{e}^{(4)}$?

Recall: the Dot Product

- ▶ The **dot product** of $\vec{u}$ and $\vec{v}$ is defined as:

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$$

where θ is the angle between $\vec{u}$ and $\vec{v}$.

- ▶ $\vec{u} \cdot \vec{v} = 0$ if and only if $\vec{u}$ and $\vec{v}$ are orthogonal

Dot Product (Coordinate Form)

- In terms of coordinate vectors:

$$\begin{aligned}\vec{u} \cdot \vec{v} &= \vec{u}^T \vec{v} \\ &= (u_1 \quad u_2 \quad \cdots \quad u_d) \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_d \end{pmatrix} \\ &= \end{aligned}$$

- This definition assumes the standard basis.

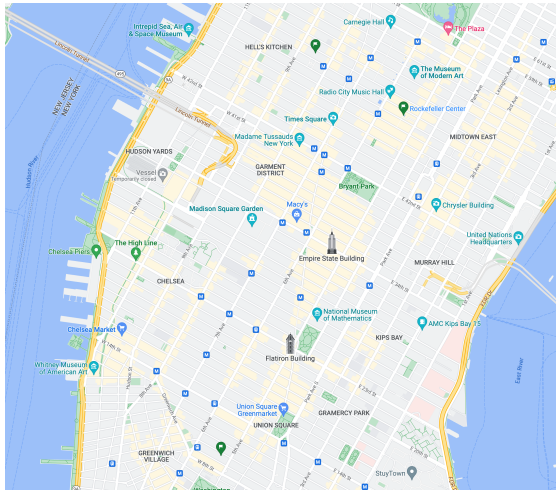
Example

$$\begin{pmatrix} 3 \\ 7 \\ 2 \\ -5 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} =$$

Other Bases

- ▶ The standard basis is not the **only** basis.
- ▶ Sometimes more convenient to use another.

Example

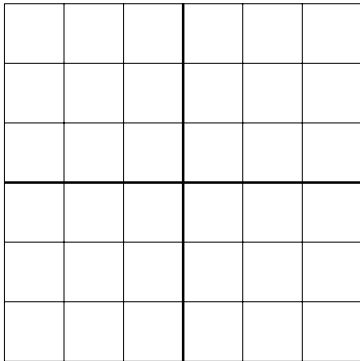


Orthonormal Bases

- ▶ **Orthonormal bases** are particularly nice.
- ▶ A set of vectors $\hat{u}^{(1)}, \dots, \hat{u}^{(d)}$ forms an **orthonormal basis** $\mathcal{U}$ for R^d if:
 - ▶ They are mutually orthogonal: $\hat{u}^{(i)} \cdot \hat{u}^{(j)} = 0$.
 - ▶ They are all unit vectors: $\|\hat{u}^{(i)}\| = 1$.

Example

$$\hat{u}^{(1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \hat{u}^{(2)} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$



Coordinate Vectors

- ▶ A vector's coordinates depend on the basis used.
- ▶ If we are using the basis $\mathcal{U} = \{\hat{u}^{(1)}, \hat{u}^{(2)}\}$, then $\vec{x} = (x_1, x_2)^T$ means $\vec{x} = x_1 \hat{u}^{(1)} + x_2 \hat{u}^{(2)}$.
- ▶ We will write $[\vec{x}]_{\mathcal{U}} = (x_1, \dots, x_d)^T$ to denote that the coordinates are with respect to the basis $\mathcal{U}$.

Exercise

Let $\hat{u}^{(1)} = \frac{1}{\sqrt{2}}(1, 1)^T$ and $\hat{u}^{(2)} = \frac{1}{\sqrt{2}}(-1, 1)^T$. Suppose $[\vec{x}]_{\mathcal{U}} = (3, -4)^T$. What is $\vec{x} \cdot \hat{u}^{(1)}$?

Exercise

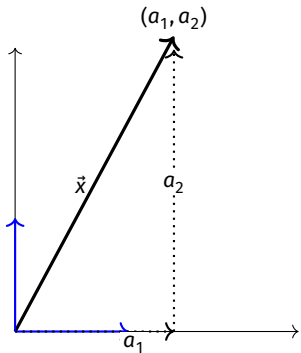
Consider $\vec{x} = (2, 2)^T$ and let $\hat{u}^{(1)} = \frac{1}{\sqrt{2}}(1, 1)^T$ and $\hat{u}^{(2)} = \frac{1}{\sqrt{2}}(-1, 1)^T$. What is $[\vec{x}]_{\mathcal{U}}$?

- ▶ A) $(0, 2\sqrt{2})^T$
- ▶ B) $(2, 2)^T$
- ▶ C) $(2\sqrt{2}, 0)^T$
- ▶ D) $(\sqrt{2}, \sqrt{2})^T$

Change of Basis

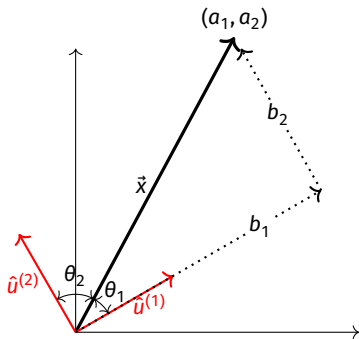
- ▶ How do we compute the coordinates of a vector in a new orthonormal basis, $\mathcal{U}$?
- ▶ Some trigonometry is involved.
- ▶ **Key Fact:** $\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \cos \theta$

Change of Basis



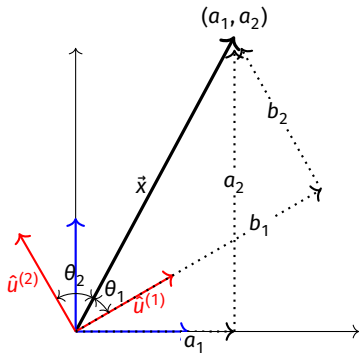
- Suppose we know $\vec{x} = (a_1, a_2)^T$ w.r.t. standard basis.
- Then $\vec{x} = a_1 \hat{e}^{(1)} + a_2 \hat{e}^{(2)}$

Change of Basis



- Want to write:
 $\vec{x} = b_1 \hat{u}^{(1)} + b_2 \hat{u}^{(2)}$
- Need to find b_1 and b_2 .

Change of Basis



- **Exercise:** Solve for b_1 , writing the answer as a dot product.
- Hint: $\cos \theta = \text{adjacent/hypotenuse}$

Change of Basis

- ▶ Let $\mathcal{U} = \{\hat{u}^{(1)}, \dots, \hat{u}^{(d)}\}$ be an orthonormal basis.
- ▶ The coordinates of $\vec{x}$ w.r.t. $\mathcal{U}$ are:

$$[\vec{x}]_{\mathcal{U}} = \begin{pmatrix} \vec{x} \cdot \hat{u}^{(1)} \\ \vec{x} \cdot \hat{u}^{(2)} \\ \vdots \\ \vec{x} \cdot \hat{u}^{(d)} \end{pmatrix}$$

Exercise

Suppose $\vec{x} = (2, 1)^T$ and let $\hat{u}^{(1)} = \frac{1}{\sqrt{2}}(1, 1)^T$ and $\hat{u}^{(2)} = \frac{1}{\sqrt{2}}(-1, 1)^T$. What is $[\vec{x}]_{\mathcal{U}}$?

- ▶ A) $\left(\frac{3\sqrt{2}}{2}, \frac{-\sqrt{2}}{2}\right)^T$
- ▶ B) $\left(\frac{\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)^T$
- ▶ C) $(2, 1)^T$
- ▶ D) $\left(\frac{3}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^T$

Exercise

Let $\vec{x} = (-1, 4)^T$ and suppose:

$$\hat{u}^{(1)} \cdot \hat{e}^{(1)} = 3$$

$$\hat{u}^{(2)} \cdot \hat{e}^{(1)} = -1$$

$$\hat{u}^{(1)} \cdot \hat{e}^{(2)} = -2$$

$$\hat{u}^{(2)} \cdot \hat{e}^{(2)} = 5$$

What is $[\vec{x}]_{\mathcal{U}}$?

- ▶ A) $(-11, 21)^T$
- ▶ B) $(21, -11)^T$
- ▶ C) $(-1, 4)^T$
- ▶ D) $(-5, 23)^T$

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Lecture 02 | Part 4

Functions of a Vector

Functions of a Vector

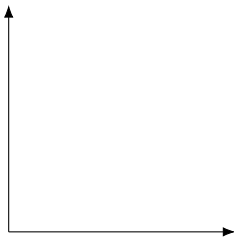
- ▶ In ML, we often work with functions of a vector:
 $f : \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$.
- ▶ Example: a prediction function, $H(\vec{x})$.
- ▶ Functions of a vector can return:
 - ▶ a number: $f : \mathbb{R}^d \rightarrow \mathbb{R}^1$
 - ▶ a vector $\vec{f} : \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$
 - ▶ something else?

Transformations

- ▶ A **transformation** $\vec{f}$ is a function that takes in a vector, and returns a vector *of the same dimensionality*.
- ▶ That is, $\vec{f} : \mathbb{R}^d \rightarrow \mathbb{R}^d$.

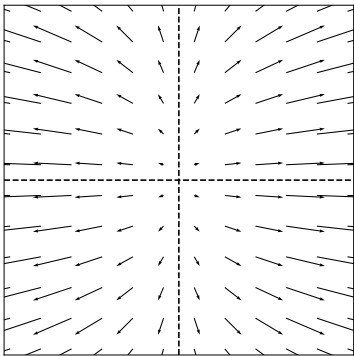
Visualizing Transformations

- ▶ A transformation is a **vector field**.
 - ▶ Assigns a vector to each point in space.
 - ▶ Example: $\vec{f}(\vec{x}) = (3x_1, x_2)^T$



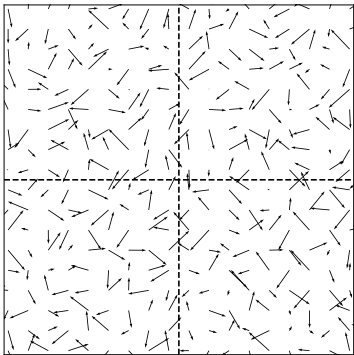
Example

► $\vec{f}(\vec{x}) = (3x_1, x_2)^T$



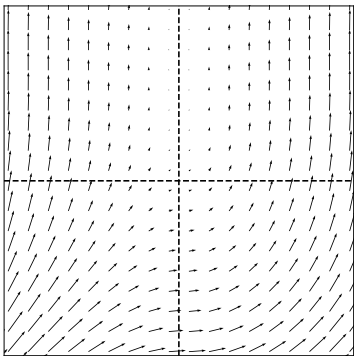
Arbitrary Transformations

- Arbitrary transformations can be quite complex.



Arbitrary Transformations

- Arbitrary transformations can be quite complex.



Linear Transformations

- ▶ Luckily, we often² work with simpler, **linear transformations**.
- ▶ A transformation f is **linear** if:

$$\vec{f}(\alpha\vec{x} + \beta\vec{y}) = \alpha\vec{f}(\vec{x}) + \beta\vec{f}(\vec{y})$$

²Sometimes just to make the math tractable!

Checking Linearity

- ▶ To check if a transformation is linear, use the definition.
- ▶ **Example:** $\vec{f}(\vec{x}) = (x_2, -x_1)^T$

Exercise

Let $\vec{f}(\vec{x}) = (x_1 + 3, x_2)$. True or False: $\vec{f}$ is a linear transformation.

Solution

- ▶ **False.** $\vec{f}$ is not a linear transformation.
- ▶ To see this, note that any linear transformation must satisfy $\vec{f}(\vec{0}) = \vec{0}$.
- ▶ However, $\vec{f}(\vec{0}) = (0 + 3, 0)^T = (3, 0)^T \neq \vec{0}$.
- ▶ Therefore, $\vec{f}$ is not linear.

Implications of Linearity

- Suppose $\vec{f}$ is a linear transformation. Then:

$$\begin{aligned}\vec{f}(\vec{x}) &= \vec{f}(x_1 \hat{e}^{(1)} + x_2 \hat{e}^{(2)}) \\ &= x_1 \vec{f}(\hat{e}^{(1)}) + x_2 \vec{f}(\hat{e}^{(2)})\end{aligned}$$

- I.e., $\vec{f}$ is **totally determined** by what it does to the basis vectors.

The **Complexity** of Arbitrary Transformations

- ▶ Suppose f is an **arbitrary** transformation.
- ▶ I tell you $\vec{f}(\hat{e}^{(1)}) = (2, 1)^T$ and $\vec{f}(\hat{e}^{(2)}) = (-3, 0)^T$.
- ▶ I tell you $\vec{x} = (x_1, x_2)^T$.
- ▶ What is $\vec{f}(\vec{x})$?

The **Simplicity** of Linear Transformations

- ▶ Suppose f is a **linear** transformation.
- ▶ I tell you $\vec{f}(\hat{e}^{(1)}) = (2, 1)^T$ and $\vec{f}(\hat{e}^{(2)}) = (-3, 0)^T$.
- ▶ I tell you $\vec{x} = (x_1, x_2)^T$.
- ▶ What is $\vec{f}(\vec{x})$?

Exercise

- ▶ Suppose f is a **linear** transformation.
 - ▶ I tell you $\vec{f}(\hat{e}^{(1)}) = (2, 1)^T$ and $\vec{f}(\hat{e}^{(2)}) = (-3, 0)^T$.
 - ▶ I tell you $\vec{x} = (3, -4)^T$.
 - ▶ What is $\vec{f}(\vec{x})$?
-
- ▶ A) $(3, 18)^T$
 - ▶ B) $(6, 3)^T$
 - ▶ C) $(-6, 3)^T$
 - ▶ D) $(18, 3)^T$

Key Fact

- ▶ Linear functions are determined **entirely** by what they do on the basis vectors.
- ▶ I.e., to tell you what f does, I only need to tell you $\vec{f}(\hat{e}^{(1)})$ and $\vec{f}(\hat{e}^{(2)})$.
- ▶ This makes the math easy!

Linear Algebra

- ▶ This is the key idea behind **linear** algebra.
- ▶ Linear algebra studies the properties of **linear** transformations.
- ▶ Non-linear transformations are **so complicated** that we can say relatively little about them.

A photograph of a formal garden, likely the gardens of Stourhead in England. The garden features a central lawn area, surrounded by meticulously manicured hedges and various plants. The background is a dense forest of tall trees. The text "Arbitrary Transformations" is overlaid on the upper right portion of the image.

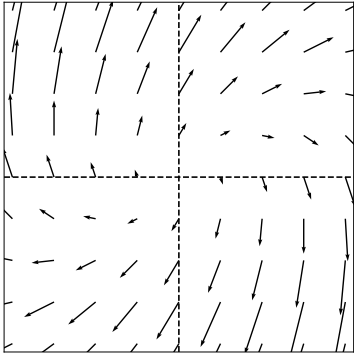
Arbitrary
Transformations

Linear
Transformations

A small, stylized logo consisting of a lowercase letter 'b' inside a square frame, located in the bottom left corner of the image.

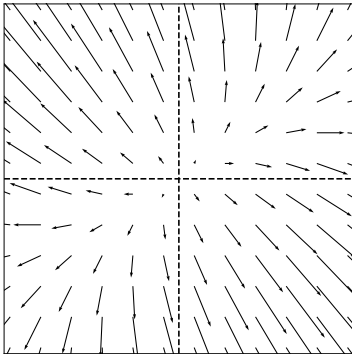
Example Linear Transformation

► $\vec{f}(\vec{x}) = (x_1 + 3x_2, -3x_1 + 5x_2)^T$



Another Example Linear Transformation

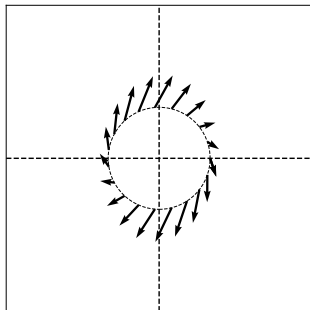
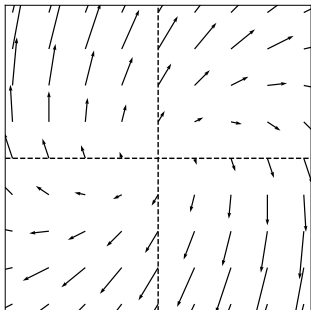
► $\vec{f}(\vec{x}) = (2x_1 - x_2, -x_1 + 3x_2)^T$

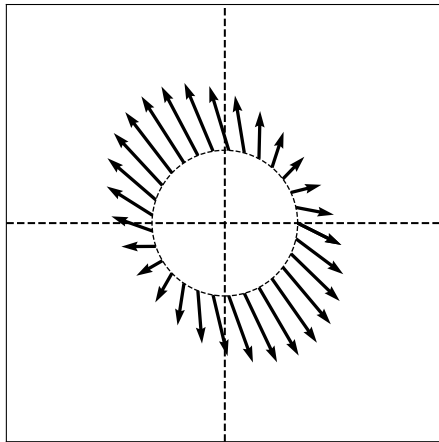
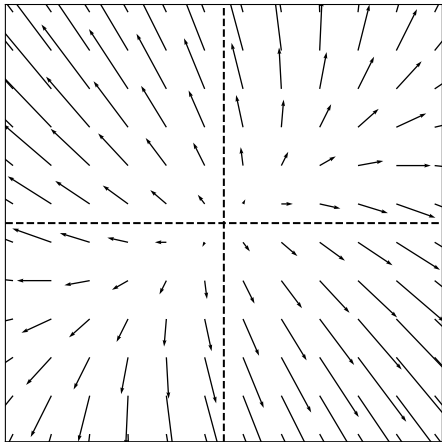


Note

- Because of linearity, along any given direction $\vec{f}$ changes only in scale.

$$\vec{f}(\lambda \hat{x}) = \lambda \vec{f}(\hat{x})$$





Linear Transformations and Bases

- We have been writing transformations in coordinate form. For example:

$$\begin{aligned}\vec{f}(\vec{x}) &= (x_1 + x_2, x_1 - x_2)^T \\ &= (x_1 + x_2)\hat{e}^{(1)} + (x_1 - x_2)\hat{e}^{(2)}\end{aligned}$$

- If we use a different basis, the formula for $\vec{f}$ **changes**:

$$\begin{aligned}[\vec{f}(\vec{x})]_{\mathcal{U}} &= (?, ?)^T \\ &= [?]\hat{u}^{(1)} + [?]\hat{u}^{(2)}\end{aligned}$$

Linear Transformations and Bases

- We know that if $\vec{x} = x_1\hat{e}^{(1)} + x_2\hat{e}^{(2)}$, then:

$$\vec{f}(\vec{x}) = (x_1 + x_2)\hat{e}^{(1)} + (x_1 - x_2)\hat{e}^{(2)}$$

- Now: if $\vec{x} = z_1\hat{u}^{(1)} + z_2\hat{u}^{(2)}$, what is:

$$\vec{f}(\vec{x}) = ?\hat{u}^{(1)} + ?\hat{u}^{(2)}$$

Key Fact

- If we use linearity:

$$\begin{aligned}f(\vec{x}) &= f(z_1 \hat{u}^{(1)} + z_2 \hat{u}^{(2)}) \\ &= z_1 f(\hat{u}^{(1)}) + z_2 f(\hat{u}^{(2)})\end{aligned}$$

- **Strategy:** to write $\vec{f}$ in the $\mathcal{U}$ basis, we just need to know what $\vec{f}$ does to $\hat{u}^{(1)}$ and $\hat{u}^{(2)}$.

Example

► Let:

$$\begin{aligned} \text{► } \vec{f}(\vec{x}) &= (x_1 + x_2, x_1 - x_2)^T \\ \text{► } \hat{u}^{(1)} &= \frac{1}{\sqrt{2}}(1, 1)^T \text{ and } \hat{u}^{(2)} = \frac{1}{\sqrt{2}}(-1, 1)^T. \end{aligned}$$

► Then:

$$\vec{f}(\hat{u}^{(1)}) = \vec{f}\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^T = (\sqrt{2}, 0)^T = \sqrt{2}\hat{e}^{(1)}$$

$$\vec{f}(\hat{u}^{(2)}) = \vec{f}\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^T = (0, -\sqrt{2})^T = -\sqrt{2}\hat{e}^{(2)}$$

► **But** we want $\vec{f}(\hat{u}^{(1)})$ and $\vec{f}(\hat{u}^{(2)})$ in terms of $\hat{u}^{(1)}$ and $\hat{u}^{(2)}$.

Example (Cont.)

- ▶ We have: $f(\hat{u}^{(1)}) = \sqrt{2}\hat{e}^{(1)}$ and $f(\hat{u}^{(2)}) = -\sqrt{2}\hat{e}^{(2)}$.
- ▶ To write $\vec{f}(\hat{u}^{(1)})$ in terms of $\hat{u}^{(1)}$ and $\hat{u}^{(2)}$, compute:

$$\begin{aligned} f(\hat{u}^{(1)}) &= (f(\hat{u}^{(1)}) \cdot \hat{u}^{(1)})\hat{u}^{(1)} + (f(\hat{u}^{(1)}) \cdot \hat{u}^{(2)})\hat{u}^{(2)} \\ &= \\ &= \end{aligned}$$

Example (Cont.)

- ▶ We have: $f(\hat{u}^{(1)}) = \sqrt{2}\hat{e}^{(1)}$ and $f(\hat{u}^{(2)}) = -\sqrt{2}\hat{e}^{(2)}$.
- ▶ To write $\vec{f}(\hat{u}^{(1)})$ in terms of $\hat{u}^{(1)}$ and $\hat{u}^{(2)}$, compute:

$$\begin{aligned} f(\hat{u}^{(1)}) &= (f(\hat{u}^{(1)}) \cdot \hat{u}^{(1)})\hat{u}^{(1)} + (f(\hat{u}^{(1)}) \cdot \hat{u}^{(2)})\hat{u}^{(2)} \\ &= \left((\sqrt{2}, 0) \cdot \frac{1}{\sqrt{2}}(1, 1) \right) \hat{u}^{(1)} + \left((\sqrt{2}, 0) \cdot \frac{1}{\sqrt{2}}(-1, 1) \right) \hat{u}^{(2)} \\ &= \end{aligned}$$

Example (Cont.)

- ▶ We have: $f(\hat{u}^{(1)}) = \sqrt{2}\hat{e}^{(1)}$ and $f(\hat{u}^{(2)}) = -\sqrt{2}\hat{e}^{(2)}$.
- ▶ To write $\vec{f}(\hat{u}^{(1)})$ in terms of $\hat{u}^{(1)}$ and $\hat{u}^{(2)}$, compute:

$$\begin{aligned}f(\hat{u}^{(1)}) &= (f(\hat{u}^{(1)}) \cdot \hat{u}^{(1)})\hat{u}^{(1)} + (f(\hat{u}^{(1)}) \cdot \hat{u}^{(2)})\hat{u}^{(2)} \\&= \left((\sqrt{2}, 0) \cdot \frac{1}{\sqrt{2}}(1, 1) \right) \hat{u}^{(1)} + \left((\sqrt{2}, 0) \cdot \frac{1}{\sqrt{2}}(-1, 1) \right) \hat{u}^{(2)} \\&= (1)\hat{u}^{(1)} + (-1)\hat{u}^{(2)} = \hat{u}^{(1)} - \hat{u}^{(2)}\end{aligned}$$

Example (Cont.)

- Similarly, for $\vec{f}(\hat{u}^{(2)})$:

$$\begin{aligned} f(\hat{u}^{(2)}) &= (f(\hat{u}^{(2)}) \cdot \hat{u}^{(1)})\hat{u}^{(1)} + (f(\hat{u}^{(2)}) \cdot \hat{u}^{(2)})\hat{u}^{(2)} \\ &= \left((0, -\sqrt{2}) \cdot \frac{1}{\sqrt{2}}(1, 1) \right) \hat{u}^{(1)} + \left((0, -\sqrt{2}) \cdot \frac{1}{\sqrt{2}}(-1, 1) \right) \hat{u}^{(2)} \\ &= (-1)\hat{u}^{(1)} + (-1)\hat{u}^{(2)} = -\hat{u}^{(1)} - \hat{u}^{(2)} \end{aligned}$$

Solution

- Putting it all together:

$$\begin{aligned}f(\vec{X}) &= f(z_1 \hat{u}^{(1)} + z_2 \hat{u}^{(2)}) \\&= z_1 f(\hat{u}^{(1)}) + z_2 f(\hat{u}^{(2)}) \\&= z_1(\hat{u}^{(1)} - \hat{u}^{(2)}) + z_2(-\hat{u}^{(1)} - \hat{u}^{(2)}) \\&= (z_1 - z_2)\hat{u}^{(1)} + (-z_1 - z_2)\hat{u}^{(2)}\end{aligned}$$

- Or, in coordinate form:

$$[f(\vec{X})]_{\mathcal{U}} = (z_1 - z_2, -z_1 - z_2)^T$$

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Representation Learning

Lecture 02 | Part 5

Matrices

Matrices?

- ▶ I thought this week was supposed to be about linear algebra... Where are the matrices?

Matrices?

- ▶ I thought this week was supposed to be about linear algebra... Where are the matrices?
- ▶ What is a matrix, anyways?

What is a matrix?

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Recall: Linear Transformations

- ▶ A **transformation** $\vec{f}(\vec{x})$ is a function which takes a vector as input and returns a vector of the same dimensionality.
- ▶ A transformation $\vec{f}$ is **linear** if

$$\vec{f}(\alpha \vec{u} + \beta \vec{v}) = \alpha \vec{f}(\vec{u}) + \beta \vec{f}(\vec{v})$$

Recall: Linear Transformations

- ▶ **Key** consequence of **linearity**: to compute $\vec{f}(\vec{x})$, only need to know what $\vec{f}$ does to basis vectors.
- ▶ Example:

$$\vec{x} = 3\hat{e}^{(1)} - 4\hat{e}^{(2)} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$$

$$\vec{f}(\hat{e}^{(1)}) = -\hat{e}^{(1)} + 3\hat{e}^{(2)}$$

$$\vec{f}(\hat{e}^{(2)}) = 2\hat{e}^{(1)}$$

$$\vec{f}(\vec{x}) =$$

Matrices

- ▶ **Idea:** Since $\vec{f}$ is defined by what it does to basis, place $\vec{f}(\hat{e}^{(1)})$, $\vec{f}(\hat{e}^{(2)})$, ... into a table as columns
- ▶ This is the **matrix** representing³ $\vec{f}$

$$\begin{aligned}\vec{f}(\hat{e}^{(1)}) &= -\hat{e}^{(1)} + 3\hat{e}^{(2)} = \begin{pmatrix} -1 \\ 3 \end{pmatrix} \\ \vec{f}(\hat{e}^{(2)}) &= 2\hat{e}^{(1)} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}\end{aligned}\qquad \begin{pmatrix} -1 & 2 \\ 3 & 0 \end{pmatrix}$$

³with respect to the standard basis $\hat{e}^{(1)}, \hat{e}^{(2)}$

Example

Write the matrix representing $\vec{f}$ with respect to the standard basis, given:

$$\vec{f}(\hat{e}^{(1)}) = (1, 4, 7)^T$$

$$\vec{f}(\hat{e}^{(2)}) = (2, 5, 7)^T$$

$$\vec{f}(\hat{e}^{(3)}) = (3, 6, 9)^T$$

Exercise

Suppose $\vec{f}$ has the matrix below:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Let $\vec{x} = (-2, 1, 3)^T$. What is $\vec{f}(\vec{x})$?

- ▶ A) $(3, 12, 21)^T$
- ▶ B) $(-2, 1, 3)^T$
- ▶ C) $(6, 15, 24)^T$
- ▶ D) $(9, 15, 21)^T$

Main Idea

A square ($n \times n$) matrix can be interpreted as a compact representation of a linear transformation $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

What is matrix multiplication?

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} \\ \\ \end{pmatrix}$$

A low-level definition

$$(A\vec{x})_i = \sum_{j=1}^n A_{ij}x_j$$

A low-level interpretation

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} = -2 \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} + 1 \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix} + 3 \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix}$$

In general...

$$\begin{pmatrix} \uparrow & \uparrow & \uparrow \\ \vec{a}^{(1)} & \vec{a}^{(2)} & \vec{a}^{(3)} \\ \downarrow & \downarrow & \downarrow \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_1 \vec{a}^{(1)} + x_2 \vec{a}^{(2)} + x_3 \vec{a}^{(3)}$$

Matrix Multiplication

$$\vec{x} = x_1 \hat{e}^{(1)} + x_2 \hat{e}^{(2)} + x_3 \hat{e}^{(3)} = (x_1, x_2, x_3)^T$$
$$\vec{f}(\vec{x}) = x_1 \vec{f}(\hat{e}^{(1)}) + x_2 \vec{f}(\hat{e}^{(2)}) + x_3 \vec{f}(\hat{e}^{(3)})$$

$$A = \begin{pmatrix} \uparrow & \uparrow & \uparrow \\ \vec{f}(\hat{e}^{(1)}) & \vec{f}(\hat{e}^{(2)}) & \vec{f}(\hat{e}^{(3)}) \\ \downarrow & \downarrow & \downarrow \end{pmatrix}$$
$$A\vec{x} = \begin{pmatrix} \uparrow & \uparrow & \uparrow \\ \vec{f}(\hat{e}^{(1)}) & \vec{f}(\hat{e}^{(2)}) & \vec{f}(\hat{e}^{(3)}) \\ \downarrow & \downarrow & \downarrow \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$
$$= x_1 \vec{f}(\hat{e}^{(1)}) + x_2 \vec{f}(\hat{e}^{(2)}) + x_3 \vec{f}(\hat{e}^{(3)})$$

Matrix Multiplication

- ▶ Matrix A represents a linear transformation $\vec{f}$
 - ▶ With respect to the standard basis
 - ▶ If we use a different basis, the matrix changes!
- ▶ Matrix multiplication $A\vec{x}$ **evaluates** $\vec{f}(\vec{x})$

What are they, *really*?

- ▶ Matrices are sometimes just tables of numbers.
- ▶ But they often have a deeper meaning.

Main Idea

A square ($n \times n$) matrix can be interpreted as a compact representation of a linear transformation $\vec{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

What's more, if A represents $\vec{f}$, then $A\vec{x} = \vec{f}(\vec{x})$; that is, multiplying by A is the same as evaluating $\vec{f}$.

Example

$$\vec{x} = 3\hat{e}^{(1)} - 4\hat{e}^{(2)} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$$

$$A =$$

$$\vec{f}(\hat{e}^{(1)}) = -\hat{e}^{(1)} + 3\hat{e}^{(2)}$$

$$\vec{f}(\hat{e}^{(2)}) = 2\hat{e}^{(1)}$$

$$\vec{f}(\vec{x}) =$$

$$A\vec{x} =$$

Note

- ▶ All of this works because we assumed $\vec{f}$ is **linear**.
- ▶ If it isn't, evaluating $\vec{f}$ isn't so simple.

Note

- ▶ All of this works because we assumed $\vec{f}$ is **linear**.
- ▶ If it isn't, evaluating $\vec{f}$ isn't so simple.
- ▶ Linear algebra = simple!

Matrices in Other Bases

- The matrix of a linear transformation wrt the **standard basis**:

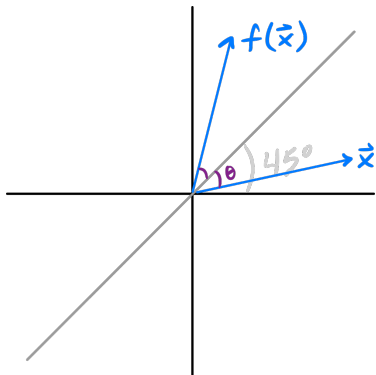
$$\begin{pmatrix} \uparrow & \uparrow & \uparrow & \\ \vec{f}(\hat{e}^{(1)}) & \vec{f}(\hat{e}^{(2)}) & \cdots & \vec{f}(\hat{e}^{(d)}) \\ \downarrow & \downarrow & \downarrow & \end{pmatrix}$$

- With respect to basis $\mathcal{U}$:

$$\begin{pmatrix} \uparrow & \uparrow & \uparrow & \\ [\vec{f}(\hat{u}^{(1)})]_{\mathcal{U}} & [\vec{f}(\hat{u}^{(2)})]_{\mathcal{U}} & \cdots & [\vec{f}(\hat{u}^{(d)})]_{\mathcal{U}} \\ \downarrow & \downarrow & \downarrow & \end{pmatrix}$$

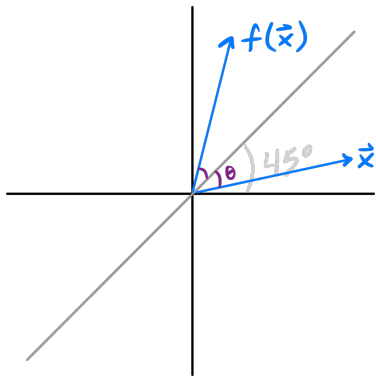
Example

- Consider the transformation $\vec{f}$ which “mirrors” a vector over the line of 45° .



- What is its matrix in the standard basis?

Example



- ▶ Let $\hat{u}^{(1)} = \frac{1}{\sqrt{2}}(1, 1)^T$
- ▶ Let $\hat{u}^{(2)} = \frac{1}{\sqrt{2}}(-1, 1)^T$
- ▶ What is $[\vec{f}(\hat{u}^{(1)})]_{\mathcal{U}}$?
- ▶ $[\vec{f}(\hat{u}^{(2)})]_{\mathcal{U}}$?
- ▶ What is the matrix?